

Chapter 2

HyperText Markup Language (HTML) Part 2

INT150 Web Technologies (2/2025)

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Topics

- What will you learn after finishing this class:
 - Presentational element
 - Inline vs. Block-level elements
 - Phrase elements
 - List elements
 - Grouping elements

Presentational elements

- are elements that indicates how text can appear in HTML.
- are **inline** (or **text-level**) **elements** that display in the flow of text by default and do not cause any line breaks.

Element	Description
b	Bold (HTML 5, Keywords)
i	Italic (HTML 5, Alternative voice)
u	Underlined
s	Strike-through (HTML 5, Incorrect text)
tt	Content is displayed in monospace font
sup	Superscript
sub	Subscript
big	Content displays one font size larger than the rest of text surrounding it
small	Content displays one font size smaller than the rest of text surrounding it

https://developer.mozilla.org/en-US/docs/Web/HTML/Inline_elements

Inline vs. Block-level elements

- **Block-level elements**

- appear on the screen as if they have a carriage return or line break before and after them.
- e.g. <p>, <h1>, , , <dl>, <pre>, <hr />
- may contain inline elements and other block-level elements.

- **Inline elements**

- can appear within sentences and do not have to appear on new lines of their own.
- e.g. , <i>, <u>, , , <sup>, <big>, <small>, <ins>, , <code>, <cite>, <kbd>, and <var>
- may contain inline elements but can not put block-level elements inside inline elements.

Inline vs. Block-level elements

`<div>`The following span is an `<span>`inline element`</span>`; its background has been colored to display both the beginning and end of the inline element's influence.`</div>`

The following span is an **inline element**; its background has been colored to display both the beginning and end of the inline element's influence.

Demo

`<div>`The following span is an `<p>` block element`</p>`; its background has been colored to display both the beginning and end of the inline element's influence.`</div>`

The following span is an

block element

; its background has been colored to display both the beginning and end of the inline element's influence.

Demo

Phrase Elements

- are displayed **similar to** the presentational elements.
- are not just **for presentational purposes**, but also **describe something** about their content.
- **Advantages** of using the phrase elements:
 - Applications such as screen readers could **add suitable intonation** the reading voice
 - **Automated programs** could **find the words** with emphasis and **extract** them out as **keywords** within a document or index.
- Appropriate use of the phrase elements adds more information to a document such as programming code, addresses

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Element	Description
em	Emphasis; display in italicized text
strong	Strong Emphasis; display in bold font
address	Address ; display in italics (different in browsers)
abbr	Abbreviations
acronym	Acronym
dfn	A special term
blockquote	Quoting text by indenting text both left and right edges
q	Add a quote within a sentence rather as an indented block
cite	Citations; display in italics
code	Programming code; display in a monospaced font
kbd	Text typed on a keyboard; display in a monospaced font
var	Programming Variables
samp	Program Output

Phrase Elements

`<p>The following description of XHTML is taken from the W3C Web site:</p>
<blockquote>XHTML 1.0 is the W3C's first Recommendation for XHTML,
following on from earlier work on HTML 4.01, HTML 4.0, HTML 3.2 and HTML
2.0. </blockquote>
<p>As Dylan Thomas said, <q>Somebody's boring me. I think it's me</q>.</p>
<p>Please press <kbd>Ctrl</kbd> + <kbd>Shift</kbd> + <kbd>R</kbd> to re-
render an MDN page.</p>`

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As Dylan Thomas said, "Somebody's boring me. I think it's me".

Please press `ctrl` + `shift` + `R` to re-render an MDN page.

Demo

List Elements

- HTML Lists allow web developers to group a set of related items in lists.
- Three types of lists
 - Unordered lists
 - Collections of items that appear in no particular orders
 - Ordered lists
 - Lists in which the sequence of the items is important.
 - Definition lists
 - List that consist of name and value pairs, including terms and definitions.

List Elements

```
<ul>
  <li>Bullet point number one</li>
  <li>Bullet point number two</li>
  <li>Bullet point number three</li>
</ul>
<ol>
  <li>Point number one</li>
  <li>Point number two</li>
  <li>Point number three</li>
</ol>
<dl>
  <dt>Unordered List</dt>
  <dd>A list of bullet points.</dd>
  <dt>Ordered List</dt>
  <dd>An ordered list of points, such as a
    numbered set of steps.</dd>
  <dt>Definition List</dt>
  <dd>A list of terms and definitions.</dd>
</dl>
```

- Bullet point number one
- Bullet point number two
- Bullet point number three

1. Point number one
2. Point number two
3. Point number three

Unordered List

A list of bullet points.

Ordered List

An ordered list of points, such as a numbered set of steps.

Definition List

A list of terms and definitions.

Demo

Attributes in : type and start

- element has attributes: **type** and **start**
 - **type**
 - To change the ordering of the list items
 - **start**
 - To specify the number that a numbered list should start at.

Type	Description	Examples
1	Arabic numbers (default)	1, 2, 3, 4, 5
A	Capital letters	A, B, C, D, E
a	Small letters	a, b, c, d, e
I	Large Roman numerals	I, II, III, IV, V
i	Small Roman numerals	i, ii, iii, iv, v

Attributes in : type and start

```
<ol type="a">  
  <li>Point number one</li>  
  <li>Point number two</li>  
  <li>Point number three</li>  
</ol>
```

a. Point number one
b. Point number two
c. Point number three

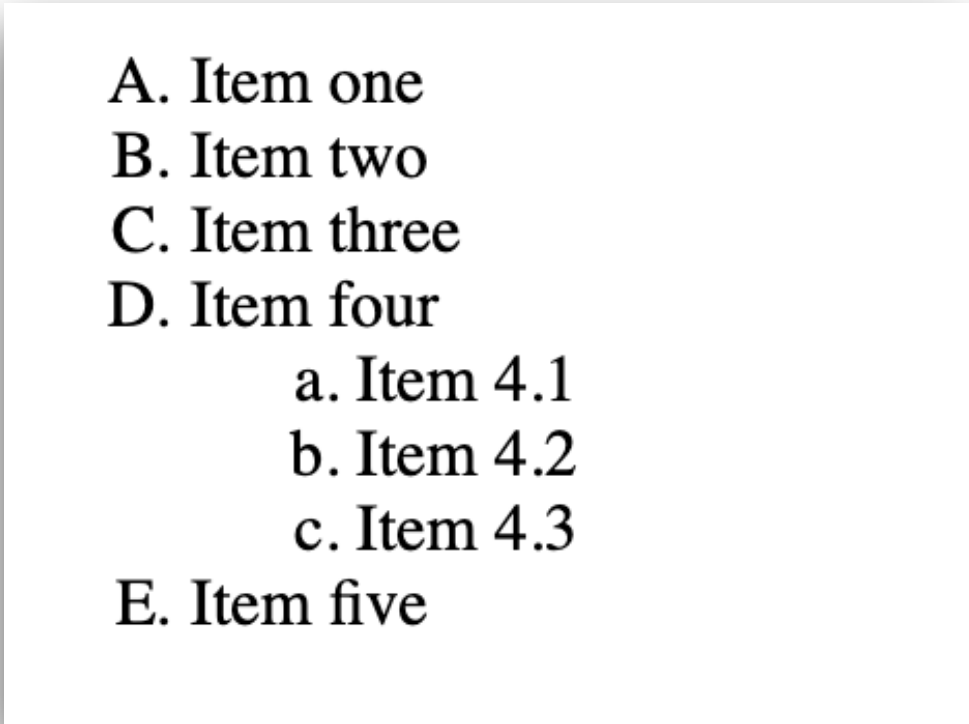
```
<ol type="i" start="5">  
  <li>Point number one</li>  
  <li>Point number two</li>  
  <li>Point number three</li>  
</ol>
```

V. Point number one
VI. Point number two
VII. Point number three

Nesting Lists

- Lists can be nested in other lists

```
<ol type="A">
  <li>Item one</li>
  <li>Item two</li>
  <li>Item three</li>
  <li>Item four
    <ol type="a">
      <li>Item 4.1</li>
      <li>Item 4.2</li>
      <li>Item 4.3</li>
    </ol>
  </li>
  <li>Item five</li>
</ol>
```



A. Item one
B. Item two
C. Item three
D. Item four
 a. Item 4.1
 b. Item 4.2
 c. Item 4.3
E. Item five

Common List Mistakes

- Improperly code when nesting lists. Any nested `<UL>` or `<OL>` must be enclosed within a parent `<LI>`.
- Misuse of list tags `<UL>` or `<OL>` to simulate paragraph indents. List tags should be used for lists, not for paragraphs.

Editing Elements

- When working on a document with others, they help to see changes that another person has made.
- They are **very similar to** a feature called **Track changes** in Microsoft Word.
- There are two elements specifically designed for **revising** and **editing text**:
 - **<ins>** element for adding text (usually shown underlined in a browser)
 - **** element for deleting some text (usually shown crossed out in a browser)

Editing Elements

`<p>My favorite color is <del>blue</del> <ins>red</ins>!</p>`

My favorite color is ~~blue~~ red!

Editing Elements: attributes

Attribute	Value	Description
title	text	Provide information as to who modified and why it was modified
cite	URL	Specifies a URL to a document that explains the reason why the text was modified
datetime	YYYY-MM-DDThh:mm:ssTZD	Specifies the date time of text was modified

Grouping Elements: <div> and

- **Grouping** (or **Generic**) elements
 - allow to **group several elements** to create sections or subsections of a page.
 - will **not affect** the **appearance** of a page.
 - are commonly **used with CSS** to **format** a style to a section of a page.
 - also allow elements on the page to be **accessed** and **manipulated** by **JavaScript**.
 - **should be used** only when **no other semantic element** is appropriate.

<https://developer.mozilla.org/en-US/docs/Web/HTML/Element/span>

Grouping Elements: <div> and

- <div> element
 - indicates a **division** of content.
 - is used to **group block-level elements**
- element
 - indicates a **word** or **phrase** for which no text-level element currently exists.
 - is used to **group inline elements only**.

id and class attributes

- For good practice, use the `id` and `class` attributes to describe the meaning and context of this group of elements.
- The `id` and `class` attributes have different purposes.
- The `id` attribute is used to assign a unique identifier to an element in the document.
- The `class` attribute classifies elements into conceptual group.
- Multiple elements may share a class name. This allows the elements of the same class to apply with a single style rule or manipulate them all with a script.

Grouping Elements: <div> and

```
<div id="news">  
  <h1>New This Week</h1>  
  <p>We've been working on...</p> <p>And last but not least,... </p>  
</div>
```

```
<ul>  
  <li>John: <span class="tel">999.8282</span></li>  
  <li>Paul: <span class="tel">888.4889</span></li>  
  <li>George: <span class="tel">888.1628</span></li>  
  <li>Ringo: <span class="tel">999.3220</span></li>  
</ul>
```

- John: 999.8282
- Paul: 888.4889
- George: 888.1628
- Ringo: 999.3220

Demo

References

- <https://developer.mozilla.org/en-US/docs/Web/HTML>
- <https://developer.mozilla.org/en-US/docs/Web/HTML/Element>
- [https://developer.mozilla.org/en-US/docs/Web/HTML/Guides/Content categories](https://developer.mozilla.org/en-US/docs/Web/HTML/Guides/Content_categories)
- <https://www.tutorialscampus.com/html5/content-models.htm>